

(3.2)

$$a) V = \text{Span}\{1, x, x^2\} \rightarrow \text{Span}\{1, x, x^2\} \stackrel{\text{base}}{=} \text{Span}\{\tilde{u}_1, \tilde{u}_2, \tilde{u}_3\}, \text{ kde } (\tilde{u}_1, \tilde{u}_2, \tilde{u}_3) \text{ je ON posporyad.$$

Procedura Gram-Schmidta ortogonalizaci:

$$i) \tilde{u}_1 = \frac{1}{\|1\|} = \frac{1}{2} \rightarrow \|1\| = \sqrt{\int_1^5 1^2 dx} = 2$$

$$ii) \tilde{u}_2 := \text{OG proj. } x \text{ na } \text{Span}\{\tilde{u}_1\} \Rightarrow \tilde{u}_2 = \left\langle \frac{1}{2}; x \right\rangle \cdot \frac{1}{2} = \int_1^5 \frac{1}{2} x^2 dx \cdot \frac{1}{2} = 6 \cdot \frac{1}{2} = 3$$

$$\tilde{u}_2 = \frac{\tilde{u}_1 - \tilde{u}_1 \tilde{u}_2}{\|\tilde{u}_1 - \tilde{u}_1 \tilde{u}_2\|} = \frac{x - 3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{\sqrt{3}}{4} (x-3)$$

$$iii) \tilde{u}_3 - \tilde{u}_3 = x^2 - \left(\int_1^5 \frac{1}{2} x^2 dx \cdot \frac{1}{2} - \int_1^5 \frac{\sqrt{3}}{4} (x-3) dx \cdot \frac{\sqrt{3}}{4} (x-3) \right) = x^2 - \frac{62}{3} \cdot \frac{1}{2} - \frac{\sqrt{3}}{4} \cdot 32 \cdot \frac{\sqrt{3}}{4} (x-3) = x^2 - \frac{31}{3} - 6x + 18 = x^2 - 6x + \frac{23}{3}$$

$$\tilde{u}_3 = \frac{\tilde{u}_3 - \tilde{u}_3}{\|\tilde{u}_3 - \tilde{u}_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})$$

$$\text{ON báze: } (\tilde{u}_1, \tilde{u}_2, \tilde{u}_3) = \left(\frac{1}{2}; \frac{\sqrt{3}}{4} (x-3); \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \right)$$

b) Máme nálezt, v ON báze vyjádření (násobek) na 3 des. místech:

$$h_1 = \langle \sin x; \frac{1}{2} \rangle = \int_1^5 \frac{1}{2} \sin x dx = 0,128$$

$$h_2 = \langle \sin x; \frac{\sqrt{3}}{4} (x-3) \rangle = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x dx = -1,493$$

$$h_3 = \langle \sin x; \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \rangle = \int_1^5 \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \sin x dx = -0,125$$

$$\text{Aproximace sinu tedy je: } h_1 \tilde{u}_1 + h_2 \tilde{u}_2 + h_3 \tilde{u}_3 = 0,128 \cdot \frac{1}{2} - 1,493 \left(\frac{\sqrt{3}}{4} (x-3) \right) - 0,125 \cdot \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) =$$

$$= -0,053x^2 - 0,332x + 1,601$$

c) Chci spočítat normu:

$$\|\sin x - (-0,053x^2 - 0,332x + 1,601)\| = \sqrt{\int_1^5 (\sin x + 0,053x^2 + 0,332x - 1,601)^2 dx} = 0,349$$